Exercise 15.7.50

Mass, Center of Mass, and Moment of Inertia to Three Decimal Places

Problem

If E is the solid E bounded by the cylinder $y^2 + z^2 = 9$ and the planes $x = 0$, $y = 3x$, and $z = 0$ in the first octant with density function $\rho(x, y, z) = x^2 + y^2$, find the following quantities, correct to three decimal places.

(a) The mass

(b) The center of mass

(c) The moment of inertia about the z-axis

The solid and the notation

The symbol E denotes the solid region. The symbol ρ denotes the density, the mass per unit volume. The symbol dV denotes the volume element. The symbols m, $(\bar{x}, \bar{y}, \bar{z})$, and I_z denote the mass, the center of mass, and the moment of inertia about the z-axis.

The cylinder $y^2 + z^2 = 9$ has its axis on the x-axis. In the first octant the cross-section in a plane $x =$ constant is a quarter disk of radius 3. The plane $y = 3x$ cuts the solid: it gives $x \le y/3$. The planes $x = 0$ and $z = 0$ are coordinate planes.

Therefore E is the set

$$E = \{(x, y, z) : y \ge 0,\ z \ge 0,\ y^2 + z^2 \le 9,\ 0 \le x \le y/3\}.$$

For a fixed point (y, z) in the quarter disk $D = \{y \ge 0,\ z \ge 0,\ y^2 + z^2 \le 9\}$, the variable x runs from 0 to $y/3$. Thus E is x-simple over D, and

$$dV = dx\, dz\, dy.$$

The order $dx\, dz\, dy$ is legal because the limits of x contain only y, and the limits of z contain only y. Figure 1 shows E; Figure 2 shows the base and the face $x = 0$.

Part (a): the mass

The mass is the integral of the density over the solid:

$$m = \iiint_E (x^2 + y^2)\, dV = \int_0^3 \int_0^{\sqrt{9-y^2}} \int_0^{y/3} (x^2 + y^2)\, dx\, dz\, dy.$$

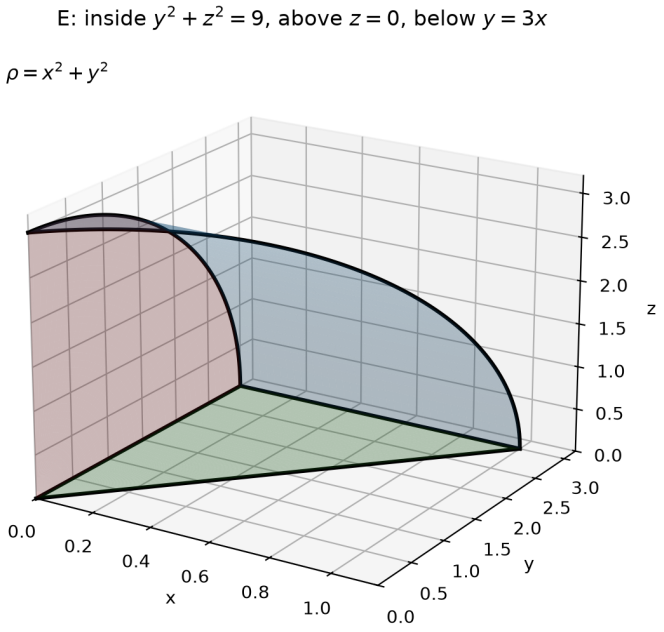


Figure 1: The solid E of Exercise 15.7.50. The blue sheet is the cylindrical wall $y^2 + z^2 = 9$. The red sheet is the quarter disk in the plane $x = 0$. The orange sheet is the planar face $y = 3x$. The green sheet is the floor $z = 0$.

The inner integral is

$$\int_0^{y/3} (x^2 + y^2)\, dx = \Big[\frac{x^3}{3} + xy^2\Big]_0^{y/3} = \frac{y^3}{81} + \frac{y^3}{3} = \frac{28y^3}{81}.$$

The integrand does not contain z, so the middle integral multiplies by the length $\sqrt{9 - y^2}$ of the column in the z-direction:

$$m = \frac{28}{81} \int_0^3 y^3 \sqrt{9 - y^2}\, dy.$$

Substitute $y = 3\sin t$, so $dy = 3\cos t\, dt$ and $\sqrt{9 - y^2} = 3\cos t$ for $0 \le t \le \pi/2$:

$$\int_0^3 y^3 \sqrt{9 - y^2}\, dy = 243 \int_0^{\pi/2} \sin^3 t \cos^2 t\, dt = 243 \cdot \frac{2}{15} = \frac{162}{5}.$$

The standard value $\int_0^{\pi/2} \sin^3 t \cos^2 t\, dt = 2/15$ follows from the beta function, or from the substitution $u = \cos t$. Therefore

$$m = \frac{28}{81} \cdot \frac{162}{5} = \frac{56}{5} = 11.2.$$

$$\boxed{m = \frac{56}{5} = 11.200}$$

Part (b): the center of mass

The center of mass is $(\bar{x}, \bar{y}, \bar{z}) = (M_{yz}/m,\ M_{xz}/m,\ M_{xy}/m)$, where $M_{yz} = \iiint x\rho\, dV$, $M_{xz} = \iiint y\rho\, dV$, and $M_{xy} = \iiint z\rho\, dV$.

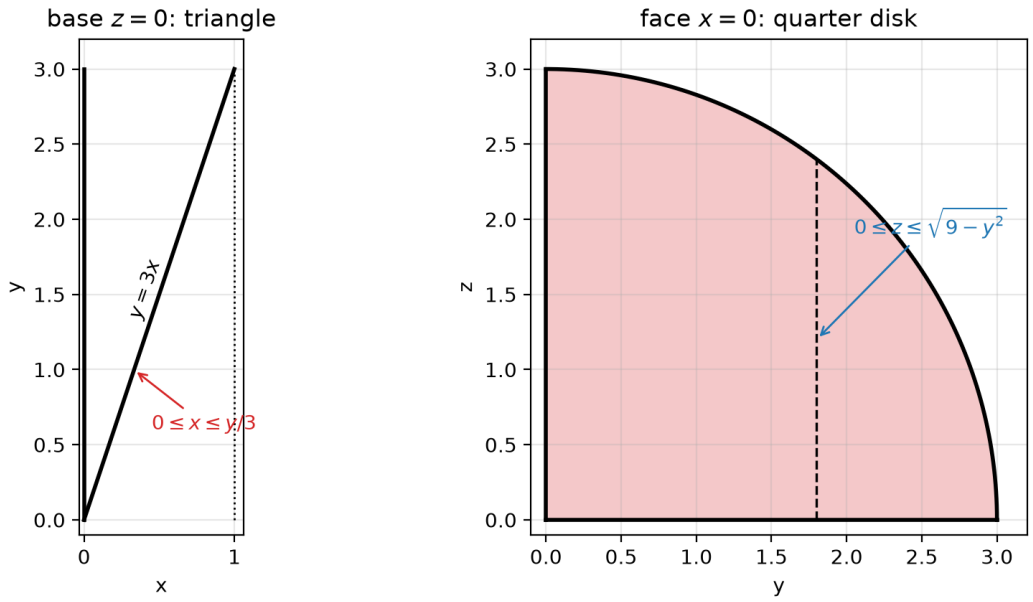


Figure 2: Left: the floor $z = 0$ is the triangle $0 \le x \le y/3$, $0 \le y \le 3$. Right: the face $x = 0$ is the quarter disk $y^2 + z^2 \le 9$; at a fixed y the height z runs from 0 to $\sqrt{9 - y^2}$.

The moment M_{yz}.

$$\int_0^{y/3} x(x^2 + y^2)\,dx = \Big[\frac{x^4}{4} + \frac{x^2y^2}{2}\Big]_0^{y/3} = \frac{y^4}{324} + \frac{y^4}{18} = \frac{19y^4}{324}.$$

Hence

$$M_{yz} = \frac{19}{324}\int_0^3 y^4\sqrt{9 - y^2}\,dy = \frac{19}{324}\cdot\frac{729\pi}{32} = \frac{171\pi}{128}.$$

The integral is $\int_0^3 y^4\sqrt{9 - y^2}\,dy = 729\int_0^{\pi/2}\sin^4 t\cos^2 t\,dt = 729\cdot\pi/32$. Therefore

$$\bar{x} = \frac{171\pi/128}{56/5} = \frac{855\pi}{7168} \approx 0.374730.$$

The moment M_{xz}.

$$\int_0^{y/3} y(x^2 + y^2)\,dx = \Big[\frac{x^3y}{3} + xy^3\Big]_0^{y/3} = \frac{y^4}{81} + \frac{y^4}{3} = \frac{28y^4}{81}.$$

Hence

$$M_{xz} = \frac{28}{81}\int_0^3 y^4\sqrt{9 - y^2}\,dy = \frac{28}{81}\cdot\frac{729\pi}{32} = \frac{63\pi}{8},$$

and

$$\bar{y} = \frac{63\pi/8}{56/5} = \frac{45\pi}{64} \approx 2.208932.$$

The moment M_{xy}. The inner integral over z is

$$\int_0^{\sqrt{9-y^2}} z\,dz = \frac{9-y^2}{2}.$$

Therefore

$$M_{xy} = \frac{1}{2}\int_0^3 (9-y^2)\Big(\int_0^{y/3}(x^2+y^2)\,dx\Big)\,dy = \frac{1}{2}\cdot\frac{28}{81}\int_0^3 y^3(9-y^2)\,dy.$$

The remaining integral is elementary:

$$\int_0^3 (9y^3-y^5)\,dy = \frac{9\cdot 81}{4} - \frac{729}{6} = \frac{729}{4} - \frac{243}{2} = \frac{243}{4}.$$

Thus

$$M_{xy} = \frac{14}{81}\cdot\frac{243}{4} = \frac{21}{2}, \qquad \bar{z} = \frac{21/2}{56/5} = \frac{15}{16} = 0.9375.$$

$$\boxed{(\bar{x},\bar{y},\bar{z}) = \Big(\frac{855\pi}{7168},\ \frac{45\pi}{64},\ \frac{15}{16}\Big) \approx (0.375,\ 2.209,\ 0.938)}$$

The last coordinate is exactly $15/16 = 0.9375$, so its three-decimal value is 0.938 with the usual round-half-up rule.

Part (c): the moment of inertia about the z-axis

The distance from the z-axis is $\sqrt{x^2+y^2}$, so

$$I_z = \iiint_E (x^2+y^2)\rho\,dV = \iiint_E (x^2+y^2)^2\,dV.$$

With the same order of integration,

$$\int_0^{y/3}(x^2+y^2)^2\,dx = \int_0^{y/3}(x^4+2x^2y^2+y^4)\,dx = \frac{y^5}{1215} + \frac{2y^5}{81} + \frac{y^5}{3} = \frac{436y^5}{1215}.$$

Hence

$$I_z = \frac{436}{1215}\int_0^3 y^5\sqrt{9-y^2}\,dy.$$

The substitution $y = 3\sin t$ gives

$$\int_0^3 y^5\sqrt{9-y^2}\,dy = 2187\int_0^{\pi/2}\sin^5 t\cos^2 t\,dt = 2187\cdot\frac{8}{105} = \frac{5832}{35}.$$

Therefore

$$I_z = \frac{436}{1215}\cdot\frac{5832}{35} = \frac{10464}{175} \approx 59.794286.$$

$$\boxed{I_z = \frac{10464}{175} \approx 59.794}$$

Quantity	Exact value	Exact (decimal)	Three decimals
m	$56/5$	11.2	11.200
$\bar{x}$	$855\pi/7168$	0.374730	0.375
$\bar{y}$	$45\pi/64$	2.208932	2.209
$\bar{z}$	$15/16$	0.937500	0.938
I_z	$10464/175$	59.794286	59.794

Table 1: Exact and rounded values for Exercise 15.7.50. The Gauss–Legendre rule on E reproduces the exact decimals; the Monte Carlo average agrees within its standard error.

Verification

Every value was computed twice: exactly with SymPy, and numerically with a Gauss–Legendre product rule and with a Monte Carlo average on E. Both routes agree. Table 1 lists the results.

Checks. The average x-coordinate must satisfy $0 \le \bar{x} \le \bar{y}/3$, because $0 \le x \le y/3$ at every point. The values $\bar{x} = 0.375$ and $\bar{y}/3 = 0.736$ satisfy this condition. The coordinate $\bar{z}$ lies between 0 and 3. With unit density the same limits give the volume

$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_0^{y/3} dx\, dz\, dy = \frac{1}{3} \iint_D y\, dA = \frac{1}{3} \cdot 9 = 3,$$

because $\iint_D y\, dA = 9$ on the quarter disk of radius 3. Exercise 15.7.18 uses the same solid. The scripts `part1_step1.py`, `part2_step1.py`, and `part3_step1.py` print `PASS`.

How to solve problems of this kind

1. **Identify the axis of the cylinder.** The equation $y^2 + z^2 = 9$ does not contain x. The cylinder is therefore parallel to the x-axis, and the natural order of integration puts x on the inside.
2. **Describe the cross-section.** For each fixed x, the cross-section is a quarter disk. For each fixed (y, z) in that disk, x runs between two surfaces. This description gives the limits directly.
3. **Read the plane as an inequality.** The plane $y = 3x$ becomes $x \le y/3$ in the first octant. The inequality gives the upper limit of the inner integral.
4. **Reduce to one-variable integrals with a trigonometric substitution.** After $y = a \sin t$, an integral $\int y^n \sqrt{a^2 - y^2}\, dy$ becomes $\sin^n t \cos^2 t$. Here $\int_0^{\pi/2} \sin^3 t \cos^2 t\, dt = 2/15$, $\int_0^{\pi/2} \sin^4 t \cos^2 t\, dt = \pi/32$, and $\int_0^{\pi/2} \sin^5 t \cos^2 t\, dt = 8/105$.
5. **Keep exact values until the last step.** Round only at the end. A chain of rounded intermediate values can change the third decimal.
6. **Check the answer against the geometry.** The center of mass must lie inside the convex hull of E. The check $\bar{x} \le \bar{y}/3$ catches many errors.
7. **Verify with two independent routes.** Compute the exact value with a computer algebra system, and the same integral numerically. Agreement to more digits than requested makes the answer reliable.